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# Defining functions
def sum(a,b):
    return a+b

def diff(a,b):
    return a-b

def prod(a,b):
    return a*b

def divd(a,b):
    if b==0:
        return "Division by 0 is not allowed"
    else:
        return a/b

def flr_divd(a,b):
    if b==0:
        return "Division by 0 is not allowed"
    else:
        return a//b

def modu(a,b):
    if b==0:
        return "Division by 0 is not allowed"
    else:
        return a%b

def expo(a,b):
    return a**b
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# main program
print("***WELCOME TO SIMPLE CALCULATOR***")

print("1. Addition")
print("2. Subtraction")
print("3. Multiplication")
print("4. Division")
print("5. Floor Division")
print("6. Modulus")
print("7. Exponentiation")
print("8. Exit\n")

# Loop for multiple calculations
ans="y"
while ans.lower()=="y":
    try:
        # Input Numbers
        a=float(input("Enter 1st No: "))
        b=float(input("Enter 2nd No: "))
        print("\n")

        # Input Operation Choices
        choice=int(input("Enter your choice: "))

        # Handling exit option
        if choice==8:
            print("Thank You for using the Simple Calculator!!!")
            break

        # Handling invalid choice
        elif choice not in range(1,8):
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        print("INVALID CHOICE\n")
        continue

    # Performing operations
    if choice==1:
        print("<<<You have selected the Addition
Calculation>>>")
        result=sum(a,b)

    elif choice==2:
        print("<<<You have selected the Subtraction
Calculation>>>")
        result=diff(a,b)

    elif choice==3:
        print("<<<You have selected the Multiplication
Calculation>>>")
        result=prod(a,b)

    elif choice==4:
        print("<<<You have selected the Division
Calculation>>>")
        result=divd(a,b)

    elif choice==5:
        print("<<<You have selected the Floor Division
Calculation>>>")
        result=f1r_divd(a,b)

    elif choice==6:
        print("<<<You have selected the Modulus
Calculation>>>")
        result=modu(a,b)
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        elif choice==7:
            print("<<<You have selected the Exponentiation  
Calculation>>>")
            result=expo(a,b)

            # Displays Result
            print("Result: ",result,"\n")

        except ValueError:
            print("INVALID INPUT. Please enter numbers only")
            continue

        # Asking whether user want to continue the calculations
        ans=input("Want to continue?(y/n): ")
        print("\n")

    else:
        print("You have discontinued the Calculation, Thank You.")
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